

Chapter 11

Areas of Polygons and Circles

Unit 4 — Area and Volume

McGraw Hill Glencoe Geometry Texas, 2015

Area Formulas Reference

Shape	Formula
Rectangle	$A = lw$
Square	$A = s^2$
Parallelogram	$A = bh$
Triangle	$A = \frac{1}{2}bh$
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$
Rhombus / Kite	$A = \frac{1}{2}d_1d_2$
Regular polygon	$A = \frac{1}{2}Pa$
Circle	$A = \pi r^2$

Postulate 11-1: Area Addition

The area of a region equals the **sum of the areas** of its non-overlapping parts.

$$A_{\text{total}} = A_1 + A_2 + \cdots + A_n$$

Theorem 11-1: Area of a Parallelogram

The area of a **parallelogram** equals the **base times the height**.

$$A = bh$$

The height h is perpendicular to the base — not the slant side!

Theorem 11-2: Area of a Triangle

The area of a **triangle** equals **half the base times the height**.

$$A = \frac{1}{2}bh$$

Alternative (Heron's Formula):

$$A = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

Theorem 11-3: Area of a Trapezoid

The area of a **trapezoid** with bases b_1 , b_2 and height h :

$$A = \frac{1}{2}h(b_1 + b_2)$$

Example: Bases 6 and 10, height 4:

$$A = \frac{1}{2}(4)(6 + 10) = \frac{1}{2}(4)(16) = 32$$

Theorem 11-4: Area of a Rhombus or Kite

The area equals **half the product of the diagonals**.

$$A = \frac{1}{2}d_1d_2$$

Example: Rhombus with diagonals 8 and 6:

$$A = \frac{1}{2}(8)(6) = 24$$

Regular Polygon — Key Terms

Term	Definition
Center	Center of the circumscribed circle
Apothem (a)	Perpendicular distance from center to a side
Central angle	$\frac{360^\circ}{n}$ for a regular n -gon

Theorem 11-5: Area of a Regular Polygon

The area of a **regular polygon** with perimeter P and apothem a :

$$A = \frac{1}{2}Pa$$

Example: Regular hexagon with side 6:

- $P = 36$, apothem $= 3\sqrt{3}$
- $A = \frac{1}{2}(36)(3\sqrt{3}) = 54\sqrt{3} \approx 93.5$

Theorem 11-6: Circumference of a Circle

$$C = 2\pi r = \pi d$$

Theorem 11-7: Area of a Circle

$$A = \pi r^2$$

Example: Radius = 5:

$$A = \pi(5)^2 = 25\pi \approx 78.5$$

Arc Length

The length of an arc with measure x° in a circle of radius r :

$$\ell = \frac{x}{360} \cdot 2\pi r$$

Area of a Sector

The area of a **sector** with central angle x° in a circle of radius r :

$$A = \frac{x}{360} \cdot \pi r^2$$

Example — Sector Area

A circle has radius 10. Find the area of a sector with central angle 72° .

$$A = \frac{72}{360} \cdot \pi(10)^2 = \frac{1}{5} \cdot 100\pi = 20\pi \approx 62.8$$

Geometric Probability — Length/Area

Geometric probability = ratio of favorable measure to total measure.

$$P = \frac{\text{favorable length or area}}{\text{total length or area}}$$

Example: Point chosen randomly inside a 10×10 square; circle of radius 3 inside. Probability of landing in circle:

$$P = \frac{\pi(3)^2}{100} = \frac{9\pi}{100} \approx 28.3\%$$

Chapter 11 — Formula Summary

Shape	Formula
Parallelogram	$A = bh$
Triangle	$A = \frac{1}{2}bh$
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$
Rhombus/Kite	$A = \frac{1}{2}d_1d_2$
Regular polygon	$A = \frac{1}{2}Pa$
Circle	$A = \pi r^2$
Sector	$A = \frac{x}{360}\pi r^2$
Arc length	$\ell = \frac{x}{360}(2\pi r)$